Eggplosion

Members:

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Why this idea?

This project was chosen because it combines **strategic gameplay with artificial intelligence decision-making** in a controlled environment. Board-game–like environments are widely used in artificial intelligence research because they provide a clear structure where agents must evaluate possible actions and select optimal strategies.

Idea Importance and Goal:

The importance of this project lies in demonstrating how **artificial intelligence agents can operate in environments that involve both strategy and uncertainty**.
The main goal of this project is to design a **grid-based chicken game environment** and implement an **AI agent** capable of selecting optimal moves using the **Depth-limited Expectiminimax algorithm**, and allow a **human player to compete against the AI agent** in real time.

Eggplosion Game Description:

This is a **two-player (Human vs. Agent), turn-based strategy game on a grid board**.

Each player controls **one chicken** that moves around the board and **lay eggs on tiles** to score points and can **place turds (obstacles)** *a limited number of turds can be placed by each chicken* to obstruct or block the opponent.

However, the board contains **hidden trapdoors**. If a chicken steps on a trapdoor, it **becomes revealed**, and the player receives a **penalty**.

The objective is to **win by laying more eggs than the opponent** before the move limit is reached.

PEAS

Performance: Maximize number of eggs, Minimize opponent eggs, Avoid trapdoors.

Environment: Grid map, Trapdoors, Eggs, Turds, Opponent Chicken.

Acuators: Keyboard (Move up, down, left, right, ‘E’ Lay eggs, ‘T’ Place turds)

Sensors: Information AI receives: Board state, Revealed tiles, Egg & Turd Locations, Chicken Positions.

ODESA

Observability: Partially Observable (because of hidden trapdoors).

Deterministic or Stochastic: Stochastic (because of hidden trapdoors).

Episodic or Sequential: Sequential.

Static or Dynamic: Static (Even though, there are hidden trapdoors, they are fixed in place and do not change randomly.

Agents: Multi-agent, Competitive (Human vs. AI).

Algorithms

Depth-limited ExpectiMinimax: Expectimax is a recursive search algorithm, it is similar to Minimax algorithm; however, while Minimax assumes an optimal adversary (worst-case scenario), Expectimax models environments where outcomes are probabilistic (in my game Eggplosion, the hidden trapdoors make the tiles unpredictable if they are safe or not).

The depth is limited because the algorithm is computationally expensive. Pruning could not be done in the case of the Expectimax as it usually requires exploring the **entire tree**. This is because a single high-value leaf can drastically change the average at a chance node.